

For use only of a Registered Medical Practitioner or a Hospital or a Laboratory

KEMOCARB SOLUTION FOR INJECTION 10MG/ML

1. NAME OF THE MEDICINAL PRODUCT

Kemocarb Solution for Injection 10 mg/ml

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

1 ml of solution for injection contains 10 mg of carboplatin.

Each 15 ml vial contains 150 mg carboplatin

Each 45 ml vial contains 450 mg carboplatin

For the full list of excipients see section 6.1.

3. PHARMACEUTICAL FORM

KEMOCARB (Carboplatin) injection is a clear, colourless to pale yellow solution. Free from visible particle.

pH- 5.0 to 7.0.

Osmolality: 200 - 300 mOsm/kg

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Carboplatin is used as initial chemotherapy of advanced ovarian carcinoma in combination with other approved chemotherapeutic agents. Carboplatin is used as palliative treatment of patients with ovarian carcinoma recurrent after prior chemotherapy, including patients who have been previously treated with cisplatin.

4.2 Posology and method of administration

Single-Agent Therapy

KEMOCARB (Carboplatin Injection BP) is given in doses of 360 mg/m² IV on day 1 every 4 weeks (alternatively see Formula Dosing). In general, however, single intermittent courses of carboplatin should not be repeated until 4 weeks after the previous course and/or until the neutrophil count is at least 2000 and the platelet count is at least 100,000.

Combination Therapy with Cyclophosphamide

In the chemotherapy of advanced ovarian cancer, carboplatin is given in doses of 300 mg/m² IV on day 1 every 4 weeks for 6 cycles (alternatively see Formula Dosing) in combination with cyclophosphamide (600 mg/m²) IV on day 1 every 4 weeks for 6 cycles.

Intermittent courses of carboplatin in combination with cyclophosphamide should not be repeated until the neutrophil count is at least 2,000 and the platelet count is at least 100,000.

Dose Adjustment Recommendations: Pretreatment platelet count and performance status are important prognostic factors for severity of myelosuppression in previously treated patients.

Dose adjustments for single agent or combination therapy shown in the table below are based on the lowest post-treatment platelet or neutrophil value.

Platelets	Neutrophils	Adjusted Dose * (From Prior Course)
>100,000	>2000	125%
50 - 100,000	500 - 2000	No Adjustment
<50,000	<500	75%

Carboplatin Injection is usually administered by an infusion lasting 15 minutes or longer. No pre- or post-treatment hydration or forced diuresis is required.

Patients with Impaired Kidney Function:

Patients with creatinine clearance values below 60 ml/min are at increased risk of severe bone marrow suppression. For the initial course of therapy, dosage modifications in the table below have been used.

Baseline Creatinine Clearance CrCL)	Recommended Dose on Day 1
41 - 59 ml/min	250 mg/m ²
16 - 40 ml/min	200 mg/m ²

Subsequent dosages should be adjusted according to the patient's tolerance based on the degree of bone marrow suppression.

Formula Dosing:

Alternatively, the use of dosing formulae, as compared to empirical dose calculation based on body surface area, allows compensation for patient variations in pretreatment renal function that might otherwise result in either underdosing or overdosing.

A simple formula for calculating dosage, based upon a patient's glomerular filtration rate (GFR in ml/min) and carboplatin target area under the concentration versus time curve (AUC in mg/ml.min), has been proposed by Calvert.

$$\text{Total dose (mg)} = \text{target (AUC)} \times (\text{GFR} + 25)$$

The target AUC of 4 to 6 mg/ml.min using single agent carboplatin appears to provide the most appropriate dose range in previously treated patients.

Geriatric Dosing:

Because renal function is often decreased in elderly patients, formula dosing of Carboplatin injection based on estimates of GFR should be used in elderly patients to provide predictable plasma Carboplatin AUCs and thereby minimize the risk of toxicity.

4.3 Contraindications

Carboplatin is contra-indicated in patients with:

- hypersensitivity to the active substance or to other platinum containing compounds
- severe myelosuppression
- bleeding tumors
- pre-existing severe renal impairment (creatinine clearance < 30 ml/min), unless in the judgement of the physician and patient, the possible benefits of treatment outweigh the risks.
- concomitant use with yellow fever vaccine (see section 4.5.)

4.4 Special warnings and precautions for use

Warnings:

Carboplatin should be administered by individuals under the supervision of a qualified physician who is experienced in the use of anti-neoplastic therapy. Blood counts as well as renal and hepatic function tests must be done regularly and the drug should be discontinued if abnormal depression of the bone marrow or abnormal renal or hepatic function is seen.

Diagnostic and treatment facilities should be readily available for management of therapy and possible complications.

Haematological toxicity

Carboplatin Infusion courses should not be repeated more frequently than monthly under normal circumstances. Leukopenia, neutropenia, and thrombocytopenia are dose-dependent and dose-limiting. Peripheral blood counts should be monitored during carboplatin injection treatment frequently and, in case of toxicity, until recovery is achieved. Median day of nadir is day 21 in patients receiving single agent carboplatin injection and day 15 in patients receiving carboplatin injection in combination with other chemotherapeutic agents.

In general, single intermittent courses of carboplatin injection should not be repeated until leukocyte, neutrophil, and platelet counts have returned to normal. Therapy should not be repeated until 4 weeks after the previous carboplatin injection course and/or until the neutrophil count is at least 2,000 cells/mm³ and the platelet count is at least 100,000 cells/mm³.

Anaemia is frequent and cumulative requiring very rarely a transfusion.

Haemolytic-uremic syndrome (HUS)

Haemolytic-uremic syndrome (HUS) is a life-threatening side effect. Carboplatin should be discontinued at the first signs of any evidence of microangiopathic haemolytic anaemia, such as rapidly falling haemoglobin with concomitant thrombocytopenia, elevation of serum bilirubin, serum creatinine, blood urea nitrogen, or LDH. Renal failure may not be reversible with discontinuation of therapy and dialysis may be required.

Severity of myelosuppression is increased in patients with prior treatment (in particular with cisplatin) and/or impaired kidney function. Initial carboplatin injection dosages in these groups of patients should be appropriately reduced (see section 4.2) and the effects carefully monitored through frequent blood counts between courses.

Carboplatin injection combination therapy with other myelosuppressive forms of treatment must be planned very carefully with respect to dosages and timing in order to minimize additive effects.

Supportive transfusional therapy may be required in patients who suffer severe myelosuppression.

Haemolytic anaemia with the presence of serologic drug-induced antibodies has been reported in patients treated with carboplatin. This event can be fatal.

Acute promyelocytic leukaemia and myelodysplastic syndrome (MDS)/ acute myeloid leukemia (AML) have been reported years after therapy with carboplatin and other antineoplastic treatments.

Myelosuppressive effects may be additive to those of concomitant chemotherapy. Patients with severe and persistent myelosuppression are at high risk of infectious complications including fatal outcomes (see section 4.8.). If any of these events occurs, carboplatin dosing should be interrupted and dose modification or discontinuation should be considered.

Hepatic and/or renal insufficiency

Renal and hepatic function impairment may be encountered with carboplatin. Very high doses of carboplatin (≥ 5 times single agent recommended dose) have resulted in severe abnormalities in hepatic and/or renal function. It is not clear whether an appropriate hydration programme might overcome effects on renal function. Dose reduction or discontinuation of therapy is required in the presence of moderate to severe alteration in renal or hepatic function test. (see section 4.8).

In patients with impaired renal function, the effect of carboplatin on the haematopoietic system is more pronounced and longer-acting than in patients with normal renal function. In this risk group, therapy with carboplatin must be performed with special caution (see section 4.2). Although no clinical evidence on compounding nephrotoxicity has been accumulated, it is recommended not to combine carboplatin with aminoglycosides or other nephrotoxic compounds (see section 4.5).

Impairment of renal function is more likely in patients who have previously experienced nephrotoxicity as a result of cisplatin therapy.

Venoocclusive liver disease

Cases of hepatic venoocclusive disease (sinusoidal obstruction syndrome) have been reported, some of which were fatal. Patients should be monitored for signs and symptoms of abnormal liver function or portal hypertension which do not obviously result from liver metastases.

Allergic reactions

As with other platinum-based drugs, allergic reactions appearing most often during perfusion may occur and necessitate discontinuation of the perfusion and an appropriate symptomatic treatment. Infrequent allergic reactions to carboplatin have been reported, e.g. erythematous rash, fever with no apparent cause or pruritus. Rarely anaphylaxis, angio-oedema and anaphylactoid reactions including bronchospasm, urticaria and facial oedema have occurred. Cross reactions, sometimes fatal, have been reported with all the platinum compounds (see section 4.3 and section 4.8).

Patients should be observed carefully for possible allergic reactions and managed with appropriate supportive therapy, including antihistamines, adrenaline and/or glucocorticoids.

The occurrence and severity and of toxicity is likely to be greater in patients who have received extensive prior treatment for their disease, have poor performance status and are advanced in years. Renal function parameters should be assessed prior to, during and after carboplatin therapy.

There have been reports of hypersensitivity reactions which progressed to Kounis syndrome (acute allergic coronary arteriospasm that can result in myocardial infarction, see section 4.8).

Neurotoxicity

Although peripheral neurologic toxicity is generally common and mild, limited to paresthesia and decrease of osteotendinous reflexes, its frequency is increased in patients older than 65 years and/or in patients previously treated with cisplatin. Monitoring and neurological examinations should be carried out at regular intervals.

Visual disturbances, including loss of vision, have been reported after the use of carboplatin injection in doses higher than those recommended in patients with renal impairment. Vision appears to recover totally or to a significant extent within weeks of stopping these high doses.

Reversible Posterior Leukoencephalopathy Syndrome (RPLS)

Cases of Reversible Posterior Leukoencephalopathy Syndrome (RPLS) have been reported in patients receiving carboplatin in combination chemotherapy. RPLS is a rare, reversible after treatment discontinuation, rapidly evolving neurological condition, which can include seizure, hypertension, headache, confusion, blindness, and other visual and neurological disturbances (see section 4.8). Diagnosis of RPLS is based upon confirmation by brain imaging, preferably MRI (Magnetic Resonance Imaging).

Ototoxicity

Auditory defects have been reported during carboplatin therapy.

Ototoxicity in children

Ototoxicity may be more pronounced in children. Cases of hearing loss with a delayed onset have been reported in paediatric patients. A long-term audiometric follow-up in this population is recommended.

Tumour lysis syndrome (TLS)

In post marketing experience tumour lysis syndrome (TLS) has been reported in patients following the use of carboplatin alone or in combination with other chemotherapeutic agents. Patient at high risk of TLS, such as patients with high proliferative rate, high tumor burden, and high sensitivity to cytotoxic agents, should be monitored closely and appropriate precaution taken.

Geriatric Use:

In studies involving combination therapy with carboplatin and cyclophosphamide, elderly patients treated with carboplatin were more likely to develop severe thrombocytopenia than younger patients. Because renal function is often decreased in the elderly, renal function should be considered when determining dosage (see section 4.2).

Vaccinations

Administration of live or live-attenuated vaccines in patients immunocompromised by chemotherapeutic agents including carboplatin, may result in serious or fatal infections. Vaccination with a live vaccine should be avoided in patients receiving carboplatin. Killed or inactivated vaccines may be administered; however, the response to such vaccines may be diminished.

Other

The carcinogenic potential of carboplatin has not been studied but compounds with similar mechanisms of action and mutagenicity have been reported to be carcinogenic (see section 5.3). Safety and effectiveness of carboplatin administration in children are not proven. Carboplatin can cause nausea and vomiting. Premedication with anti-emetics has been reported to be useful in reducing the incidence and intensity of these effects.

Aluminium containing equipment should not be used during preparation and administration of carboplatin (see section 6.2). Aluminum reacts with carboplatin injection causing precipitate formation and/or loss of potency.

Appropriate measures to prevent pregnancy should be taken during and for at least 6 months after treatment. Men should also take contraceptive measures during and for at least 3 months after treatment, as chromosomes in human spermatozoa may be destroyed due to the mutagenic potential of carboplatin.

If childbearing is desired, a consultation about sperm preservation is recommended prior to initiation of therapy. Pregnant women should avoid handling carboplatin.

4.5 Interaction with other medicinal products and other forms of interaction

When combining carboplatin with other myelosuppressive compounds or radiation therapy, the myelosuppressive effect of carboplatin and/or the other compounds may be more pronounced. Patients receiving concomitant therapy with other nephrotoxic agents are likely to experience more severe and prolonged myelotoxicity due to decreased renal clearance of carboplatin.

Concomitant use contraindicated

- Yellow fever vaccine: risk of generalised vaccinal disease mortal ((see section 4.3.).

Concomitant use not recommended

- Live attenuated vaccines (except yellow fever): risk of systemic, possible fatal disease. This risk is increased in subjects who are already immunosuppressed by their underlying disease. Use an inactivated vaccine where this exist (poliomyelitis).
- Phenytoin, fosphenytoin risk of exacerbation of convulsions resulting from the decrease of phenytoin digestive absorption by the cytotoxic drug or risk of toxicity enhancement or lose of efficacy of the cytotoxic drug due to increased hepatic metabolism by phenytoin.

Concomitant use to be taken into consideration

- Chelating agents - decreasing effect of carboplatin
- Cyclosporin (and by extrapolation tacrolimus and sirolimus): Excessive immunosuppression with risk of lymphoproliferation.
- Aminoglycosides: The concomitant use of carboplatine with aminoglycosides antibiotics should be taken into account due to the cumulative nephrotoxicity and ear toxicity, particularly in patient with severe renal impairment.
- Loop diuretics: The concomitant use of carboplatine with loop diuretic should be taken into account due to the cumulative nephrotoxicity and ear toxicity.
- Due to the increase of thrombotic risk in case of tumoral diseases, the use of anticoagulative treatment is frequent. The high intra-individual variability of the coagulability during diseases, and the eventuality of interaction between oral anticoagulants and anticancer chemotherapy, require, if it is decided to treat the patient with VKA, to increase frequency of the control of the INR monitoring. Caution and more frequent INR monitoring is recommended at concomitant treatment of warfarin with Carboplatin, as increased INR has been reported.

Carboplatin may interact with aluminium to form a black precipitate. Needles, syringes, catheters or IV administration sets that contain aluminium parts which may come into contact with carboplatin, should not be used for the preparation or administration of the drug.

4.6 Fertility, pregnancy and lactation

Pregnancy

Carboplatin injection can cause fetal harm when administered to a pregnant woman. Carboplatin injection has been shown to be embryotoxic and teratogenic in rats receiving the drug during organogenesis. No controlled studies in pregnant women have been conducted.

If this drug is used during pregnancy, or if the patient becomes pregnant while treated with this drug, the patient should be apprised of the potential hazard to the fetus. Women with child-bearing potential have to use effective contraception during and for at least 6 months after treatment.

Breast-feeding:

It is not known whether carboplatin injection is excreted in human milk.

If treatment becomes necessary during the lactation period, breastfeeding must be stopped.

Fertility

Gonadal suppression resulting in amenorrhea or azospermia may occur in patients receiving antineoplastic therapy. These effects appear to be related to dose and length of therapy and may be irreversible. Prediction of the degree of testicular or ovarian function impairment is complicated by the common use of combinations of several antineoplastics, which makes it difficult to assess the effects of individual agents.

Carboplatin is genotoxic. Men of sexually mature age treated with Carboplatin are recommended not to father a child during treatment and for at least 3 months afterwards and to ask advice about spermatoc preservation prior to initiation of the therapy because of the possibility of irreversible infertility due to therapy with carboplatin.

Women with child-bearing potential

Women with child-bearing potential should be advised to avoid becoming pregnant. Carboplatin must not be used in pregnant women or women of childbearing potential who might become pregnant unless the potential benefits to the mother outweigh the possible risks to the foetus.

If this drug is used during pregnancy, or if the patient becomes pregnant while treated with this drug, the patient should be apprised of the potential hazard to the foetus.

4.7 Effects on ability to drive and use machines

No studies on the effects on the ability to drive and use machines have been performed. However carboplatin may cause nausea, vomiting, vision abnormalities and ototoxicity; therefore, patients must be warned on the potential effect of these events on the ability to drive or to use machines.

4.8 Undesirable effects

The frequency of adverse reactions reported is based on a cumulative database of 1,893 patients receiving single agent carboplatin injection and post-marketing experience.

The list is presented by system organ class, MedDRA preferred term, and frequency using the following frequency categories:

Very common ($\geq 1/10$)

Common ($\geq 1/100$ to $< 1/10$)

Uncommon ($\geq 1/1,000$ to $\leq 1/100$)

Rare ($\geq 1/10,000$ to $\leq 1/1,000$)

Very rare ($< 1/10,000$), not known (cannot be estimated from the available data).

System Organ Class	Frequency	MedDRA Term
Infections and infestations	Common	Infections*
	Not known	Pneumonia
Neoplasms, benign, malignant and unspecified (incl cysts and polyps)	Uncommon	Treatment related secondary malignancy
Blood and lymphatic system disorders	Very common	Thrombocytopenia, neutropenia, leukopenia, anaemia
	Common	Haemorrhage*
	Rare	Febrile neutropenia
	Not known	Hemolytic-uraemic syndrome (HUS), bone marrow failure, haemolytic anaemia (sometimes fatal)
Immune system	Common	Hypersensitivity, anaphylactoid type reaction
	Rare	Anaphylaxis, anaphylactic shock, angio-oedema
Metabolism and nutrition disorders	Very common	Hyperuricaemia
	Rare	Hyponatraemia, anorexia
	Not known	Dehydration, Tumor lysis syndrome
Nervous system disorders	Common	Neuropathy peripheral, paraesthesia, decrease of osteotendinous reflexes, sensory disturbance, dysgeusia
	Very rare	Cerebrovascular accident*
	Not known	Reversible Posterior Leukoencephalopathy Syndrome (RPLS)

System Organ Class	Frequency	MedDRA Term
Eye disorders	Common	Visual disturbance (incl. rare cases of loss of vision)
	Rare	Optic neuritis
Ear and labyrinth disorders	Very common	Subclinical decrease in hearing acuity, consisting of high-frequency (4000-8000 Hz) hearing loss
	Common	Tinnitus, ototoxicity
Cardiac disorders	Common	Cardiovascular disorder*
	Very rare	Cardiac failure*
	Not known	Kounis syndrome
Vascular disorders	Very rare	Embolism*, hypertension, hypotension
Respiratory, thoracic and mediastinal disorders	Common	Respiratory disorder, Interstitial lung disease, bronchospasm
Gastrointestinal disorders	Very common	Vomiting, nausea, abdominal pain
	Common	Diarrhoea, constipation, mucous membrane disorder
	Not known	Stomatitis, pancreatitis
Hepatobiliary disorders	Rare	Severe hepatic dysfunction
Skin and subcutaneous tissue disorders	Common	Alopecia, skin disorder, urticaria, rash erythematous, pruritus
Musculoskeletal and connective tissue disorders	Common	Musculoskeletal disorder
Renal and urinary disorders	Common	Urogenital disorder
General disorders and administration site conditions	Very common	Asthenia
	Common	flu-like syndrome
	Uncommon	Fever and chills without evidence of infection, injection site necrosis, injection site reaction, injection site extravasation, injection site erythema, malaise
Investigations	Very Common	Creatinine renal clearance decreased, blood urea increased, blood alkaline phosphatase increased, aspartate aminotransferase increased, liver function test abnormal, blood sodium decreased, blood potassium decreased, blood calcium decreased, blood magnesium decreased.
	Common	Blood bilirubin increased, blood creatinine increased, blood uric acid increased

* Fatal in <1%, fatal cardiovascular events in <1% included cardiac failure, embolism, and cerebrovascular accident combined.

Neoplasms benign, malignant and unspecified (including cysts and polyps)

Secondary malignancies (including promyelocytic leukaemia which occurred 6 years after monotherapy with carboplatin and preceding irradiation) have been reported following

administration of carboplatin as a single agent or in combination therapy (causal relationship not established).

Haematologic

Myelosuppression is the dose-limiting toxicity of carboplatin injection. In patients with normal baseline values, thrombocytopenia with platelet counts below 50,000/mm occurs in 25% of patients, neutropenia with granulocyte counts below 1,000/mm³ in 18% of patients, and leukopenia with WBC counts below 2,000/mm³ in 14% of patients. The nadir usually occurs on day 21.

Myelosuppression can be worsened by combination of carboplatin injection with other myelosuppressive compounds or forms of treatment.

Myelotoxicity is more severe in previously treated patients, in particular in patients previously treated with cisplatin and in patients with impaired kidney function. Patients with poor performance status have also experienced increased leukopenia and thrombocytopenia. These effects, although usually reversible, have resulted in infectious and hemorrhagic complications in 4% and 5% of patients given carboplatin injection, respectively. These complications have led to death in less than 1% of patients.

Anaemia with haemoglobin values below 8 g/dl has been observed in 15% of patients with normal baseline values. The incidence of anaemia is increased with increasing exposure to carboplatin injection.

Immune system disorders

Allergic Reactions:

Anaphylactic-type reactions, sometimes fatal, may occur most often in the minutes following injection of the product: facial oedema, dyspnoea, tachycardia, low blood pressure, urticaria, anaphylactic shock, bronchospasm (see section 4.4)

These reactions are similar to those observed after administration of other platinum containing compounds and should be managed with appropriate supportive therapy.

Metabolism and nutrition disorders

Electrolytes:

Decreases in serum sodium, potassium, calcium, and magnesium occur in 29%, 20%, 22%, and 29% of patients, respectively. In particular, cases of early hyponatraemia have been reported. The electrolyte losses are minor and mostly take a course without any clinical symptoms.

Neurologic

Peripheral neuropathy (mainly paresthesias and decrease of osteotendinous reflexes) has occurred in 4% of patients administered carboplatin injection. Patients older than 65 years and patients previously treated with cisplatin, as well as those receiving prolonged treatment with carboplatin injection, appear to be at increased risk.

Clinically significant sensory disturbances (ie, visual disturbances and taste modifications) have occurred in 1% of patients.

The overall frequency of neurologic side effects seems to be increased in patients receiving carboplatin injection in combination. This may also be related to longer cumulative exposure.

Ear and labyrinth disorders

Ototoxicity: Auditory defects out of the speech range with impairments in the high-frequency range (4,000-8,000 Hz) were found in serial audiometric investigations with a frequency of 15%. Very rare cases of hypoacusia have been reported.

In patients with a hearing organ predamaged due to cisplatin, a further exacerbation in the hearing function sometimes occurs during treatment with carboplatin.

Gastrointestinal disorders

Vomiting occurs in 65% of patients, in one-third of whom it is severe. Nausea occurs in an additional 15%. Previously treated patients (in particular patients previously treated with cisplatin) appear to be more prone to vomiting.

Nausea and vomiting are generally delayed until 6 to 12 hours after administration of carboplatin.

These effects usually disappear within 24 hours after treatment and are generally responsive to or prevented by antiemetic medication. Vomiting is more likely when carboplatin injection is given in combination with other emetogenic compounds.

The other gastro-intestinal complaints corresponded to pain in 8% of patients, diarrhoea, and constipation in 6 % of patients.

Hepatobiliary disorders

Modification of liver function in patients with normal baseline values was observed, including elevation of total bilirubin in 5%, SGOT in 15%, and alkaline phosphatase in 24% of patients. These modifications were generally mild and reversible in about one-half the patients.

In a limited series of patients receiving very high dosages of carboplatin injection and autologous bone marrow transplantation, severe elevation of liver function tests has occurred.

Rare: Cases of an acute, fulminant liver cell necrosis occurred after high-dosed administration of carboplatin.

Renal and urinary disorders

When given in usual doses, development of abnormal renal function has been uncommon, despite the fact that carboplatin injection has been administered without high-volume fluid hydration and/or forced diuresis. Elevation of serum creatinine occurs in 6% of patients, elevation of blood urea nitrogen in 14%, and of uric acid in 5% of patients. These are usually mild and are reversible in about one-half the patients. Creatinine clearance has proven to be the most sensitive renal function measure in patients receiving carboplatin injection. Twenty-seven percent (27%) of patients who have a baseline value of 60 ml/min or greater, experience a reduction in creatinine clearance during carboplatin injection therapy. The incidence and severity of nephrotoxicity may increase in patients who have impaired kidney function before carboplatin treatment. It is not clear whether an appropriate hydration programme might overcome such an effect, but dosage reduction or discontinuation of therapy is required in the presence of moderate alteration of renal function (creatinine clearance 41-59 ml/min) or severe renal impairment (creatinine clearance 21-40 ml/min). Carboplatin is contraindicated in patients with a creatinine clearance at or below 20 ml/min.

Other undesirable effects

Secondary acute malignancies after cytostatic combination therapies containing carboplatin have been reported.

Alopecia, fever and chills, mucositis, asthenia, malaise as well as dysgeusia have occasionally been observed.

Isolated cases of haemolytic-uraemic syndrome have been reported.

Isolated cases of cardiovascular incidents (cardiac insufficiency, embolism) as well as isolated cases of cerebrovascular accidents have been reported.

Cases of hypertension have been reported.

Local reactions

Reactions at the site of injection (burning, pain, reddening, swelling, urticaria, necrosis in connection with extravasation) have been reported.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product. Healthcare professionals are asked to report any suspected adverse reactions via the national reporting system.

4.9 Overdose

Symptoms of overdose

Carboplatin was administered in Phase I studies at a dosage of up to 1600 mg/m² I.V. per course. At this dosage, life-threatening haematological side effects with granulocytopenia, thrombocytopenia and anaemia were observed. The granulocyte, thrombocyte and haemoglobin nadir were observed between days 9-25 (median: days 12-17). The granulocytes had reached values of $\geq 500/\mu\text{l}$ after 8-14 days (median: 11) and the thrombocytes values of $\geq 25.000/\mu\text{l}$ after 3-8 days (median: 7).

The following non-haematological side effects also occurred: renal function disturbances with a 50% drop in the glomerular filtration rate, neuropathy, ototoxicity, sight loss, hyperbilirubinaemia, mucositis, diarrhoea, nausea and vomiting with headache, erythema, and severe infection. In the majority of cases, hearing disturbances were transient and reversible.

Treatment of overdose

There is no known antidote for carboplatin over dosage. The anticipated complications of over dosage would be related to myelosuppression as well as impairment of hepatic and renal and auditory function. Bone marrow transplantation and transfusions (thrombocytes, blood) can be effective measures of managing haematological side effects. Use of higher than recommended doses of carboplatin injection has been associated with loss of vision (see section 4.4).

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Antineoplastic agents, platinum compounds, ATC code: L01XA02. Carboplatin is an antineoplastic agent. Its activity has been demonstrated against several murine and human cell lines.

Carboplatin exhibited comparable activity to cisplatin against a wide range of tumours regardless of implant site.

Mechanism of action

Alkaline elution techniques and DNA binding studies have demonstrated the qualitatively similar modes of action of carboplatin and cisplatin. Carboplatin, like cisplatin, induces changes in the superhelical conformation of DNA, which is consistent with a "DNA shortening effect".

Paediatric population

Paediatric patients: safety and efficacy in children have not been established (see section 4.2, 4.4 and 5.2).

5.2 Pharmacokinetic properties

Distribution

Repeated dosing during four consecutive days did not produce an accumulation of platinum in plasma.

Biotransformation

Following the administration of carboplatin reported values for the terminal elimination of half-lives of free ultra-filterable platinum and carboplatin in man are approximately 6 hours and 1.5 hours respectively. During the initial phase, most of the free ultra-filterable platinum is present as carboplatin. The terminal half-life for total plasma platinum is 24 hours. Approximately 87% of plasma platinum is protein bound within 24 hours following administration.

Elimination

Carboplatin is excreted primarily in the urine, with recovery of approximately 70% of the administered platinum within 24 hours. Most of the drug is excreted in the first 6 hours. Total body and renal clearances of free ultra-filterable platinum correlate with the rate of glomerular filtration but not tubular secretion.

Carboplatin clearance has been reported to vary by 3- to 4- fold in paediatric patients (see section 4.2 and 4.4). As for adult patients, literature data suggest that renal function may contribute to the variation in carboplatin clearance.

Linearity/non-linearity

Following administration of carboplatin in man, linear relationships exist between dose and plasma concentrations of total and free ultra-filterable platinum. The area under the plasma concentration versus time curve for total platinum also shows a linear relationship with the dose when creatinine clearance ≥ 60 ml/min.

5.3 Preclinical safety data

Carboplatin has been shown to be embryotoxic and teratogenic in rats. It is mutagenic *in vivo* and *in vitro* and although the carcinogenic potential of carboplatin has not been studied, compounds with similar mechanisms of action and mutagenicity have been reported to be carcinogenic.

Toxicity studies have shown that extravasal administration of carboplatin causes tissue necrosis.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Water for Injections

6.2 Incompatibilities

This medicinal product must not be mixed with other medicinal product except those mentioned in section 6.6.

Carboplatin may interact with aluminium to form a black precipitate. Needles, syringes, catheters or IV administration sets containing aluminium parts that may come into contact with carboplatin should not be used for preparation or administration of carboplatin. Precipitation can lead to a reduction of the antineoplastic activity.

6.3 Shelf life

Unopened: 2 years

After dilution

KEMOCARB (Carboplatin Injection BP) should be diluted to concentrations as low as 0.5 mg/ml with 5 % Dextrose in Water (D5W) or 0.9 % Sodium Chloride Injection, USP. When prepared as directed, KEMOCARB aqueous solutions are stable for 8 hours at room temperature (25°C). Since no antibacterial preservative is contained in the formulation, it is recommended that Carboplatin aqueous solutions be discarded 8 hours after dilution.

6.4 Special precautions for storage

Unopened vials: Store below 30°C. Do not freeze. Protect from light.

After dilution: For storage conditions of the diluted medicinal product, see section 6.3.

6.5 Nature and contents of container

Clear glass vial with rubber closure with aluminium flip-off seal, containing 150 mg of carboplatin, as sterile injection.

Clear glass vial with rubber closure with aluminium flip-off seal, containing 450mg of carboplatin, as sterile injection.

Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

This product is for single use only. Any unused infusion solution should be discarded.

Instruction for dilution

The product must be diluted prior to infusion, with 5 % Dextrose in Water (D5W) or 0.9 % Sodium Chloride for Injection, to concentrates as low as 0.5 mg/ml (500 micrograms/ml).

The solution is to be inspected visually for particulate matter and discoloration prior to administration. The solution should only be used if the solution is clear and free from particles.

Handling and Disposal

Procedures for proper handling and disposal of anti-cancer drugs should be considered. Use of 5 % sodium hypochlorite solution is recommended as neutralizing agent in cases of spills or leak of this solution. Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MANUFACTURER

Fresenius Kabi Oncology Limited
Village-Kishanpura,
Tehsil-Nalagarh,
Distt. Solan [H.P.] –174101

8. PRODUCT REGISTRATION HOLDER/IMPORTER

Fresenius Kabi Malaysia Sdn. Bhd.
#3-1 & 3-2, Axis Technology Centre,
Lot 13, Jalan 51A/225, 46100 Petaling Jaya,
Selangor Darul Ehsan, Malaysia.

9. DATE OF REVISION OF THE TEXT

Sep 2025